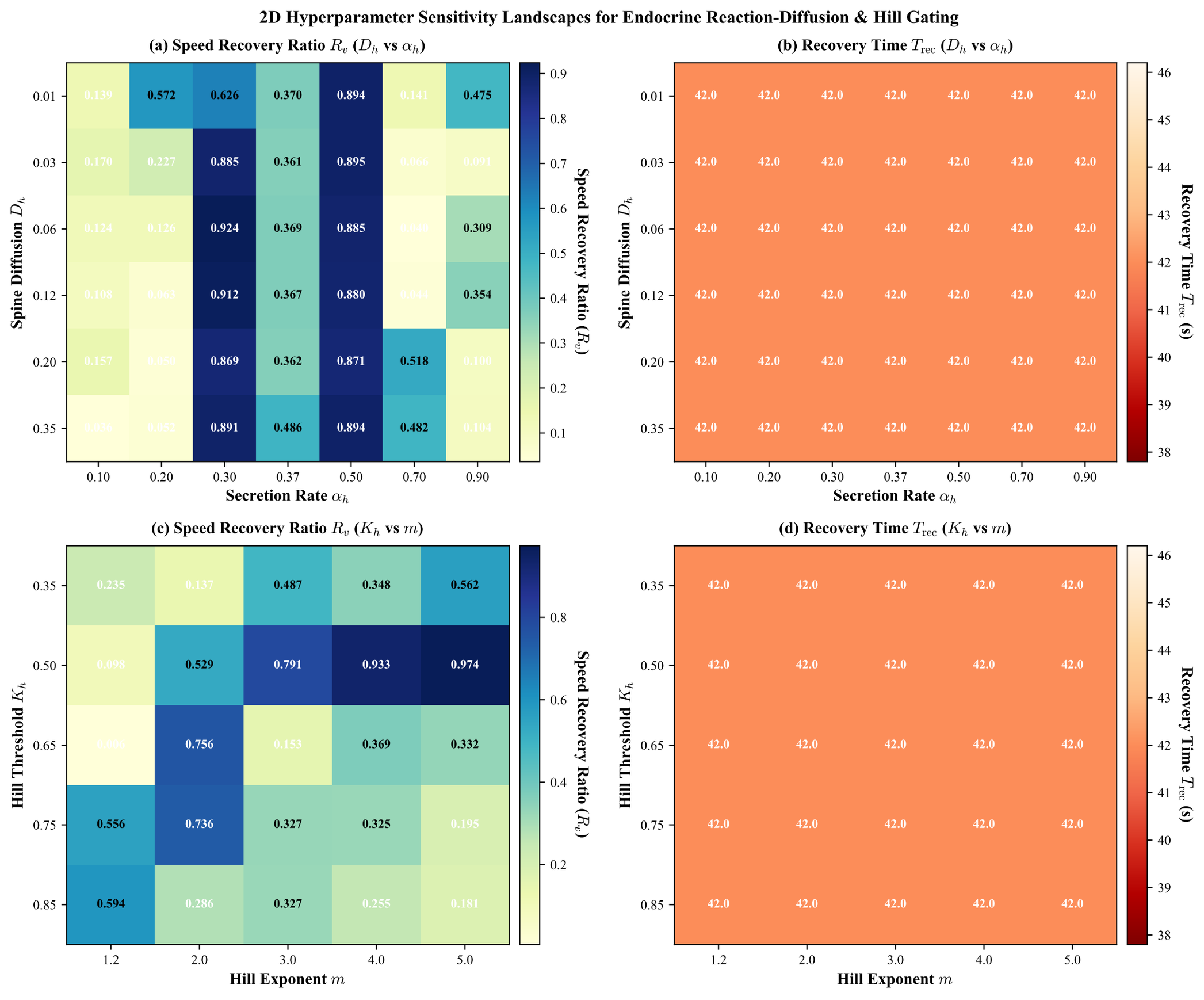

2D Hyperparameter Sensitivity Landscapes for Endocrine Reaction-Diffusion & Hill Gating
(a) Speed Recovery Ratio R_v (D_h vs α_h)
Spine Diffusion D_h
0.01
0.139 0.572 0.626 0.370 0.894 0.141 0.475
0.03
0.170 0.227 0.885 0.361 0.895 0.066 0.091
0.06
0.124 0.126 0.924 0.369 0.885 0.040 0.309
0.12
0.108 0.063 0.912 0.367 0.880 0.044 0.354
0.20
0.157 0.050 0.869 0.362 0.871 0.518 0.100
0.35
0.036 0.052 0.891 0.486 0.894 0.482 0.104
0.10 0.20 0.30 0.37 0.50 0.70 0.90
Secretion Rate α_h
Speed Recovery Ratio (R_v)
0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2 0.1
(b) Recovery Time T_{rec} (D_h vs α_h)
Spine Diffusion D_h
0.01 0.03 0.06 0.12 0.20 0.35
42.0 42.0 42.0 42.0 42.0 42.0 42.0
0.10 0.20 0.30 0.37 0.50 0.70 0.90
Secretion Rate α_h
Recovery Time T_{rec} (s)
46 45 44 43 42 41 40 39 38
(c) Speed Recovery Ratio R_v (K_h vs m)
Hill Threshold K_h
0.35
0.235 0.137 0.487 0.348 0.562
0.50
0.098 0.529 0.791 0.933 0.974
0.65
0.006 0.756 0.153 0.369 0.332
0.75
0.556 0.736 0.327 0.325 0.195
0.85
0.594 0.286 0.327 0.255 0.181
1.2 2.0 3.0 4.0 5.0
Hill Exponent m
Speed Recovery Ratio (R_v)
0.8 0.6 0.4 0.2
(d) Recovery Time T_{rec} (K_h vs m)
Hill Threshold K_h
0.35 0.50 0.65 0.75 0.85
42.0 42.0 42.0 42.0 42.0
1.2 2.0 3.0 4.0 5.0
Hill Exponent m
Recovery Time T_{rec} (s)
46 45 44 43 42 41 40 39 38